Windowed multifractal and autocorrelation analysis of seismicity

Counts, energy proxy, and temporal evolution diagnostics

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Goals of the analysis

- Quantify temporal organization in the seismic catalog using:
  - autocorrelation function (ACF),
  - MFDFA fluctuation functions $F_q(s)$,
  - generalized Hurst exponents $h(q)$,
  - multifractal spectrum $f(\alpha)$.
- Compare behavior across different seismic observables:
  - event counts,
  - energy proxy $\sum 10^{1.5M}$,
  - subcatalogs above a magnitude threshold.
- Track the *temporal evolution* of scaling via windowed $h(q)$, $\Delta h(t)$, and $\Delta\alpha(t)$.

Workflow and key commands

Main production command

Example

```
python analyze_seismic_multifractal_improved.py
--catalog eq_data_earthquake_reviewed_mag4.csv
--out results_seismic_full
--bin 1D --series energy --log1p
--qmin -10 --qmax 10 --qstep 1
--fit_smin 121 --fit_smax 4113
--maxlag 365 --acf_shuffle_band --acf_shuffle_count 200
--n_shuffles 50
--window_size_bins 3650 --window_step_bins 365
--window_q_values 0 1 2 3 4 5
```

- Daily energy proxy, 10-year windows, 1-year step.
- Conservative and stable q-range: $[-10, 10]$.
- Windowed outputs support long-term evolution analysis.

Methodological summary

MFDFA pipeline

1. Build an evenly sampled time series from the catalog.
2. Optionally apply $\log(1+x)$ and standardization.
3. Compute the profile $Y(i) = \sum_{k=1}^{i}(x_k - \bar{x})$.
4. Estimate fluctuation functions $F_q(s) \sim s^{h(q)}$.
5. Derive

$$\tau(q) = q\,h(q) - 1, \qquad \alpha = \frac{d\tau}{dq}, \qquad f(\alpha) = q\alpha - \tau(q).$$

Windowed diagnostics

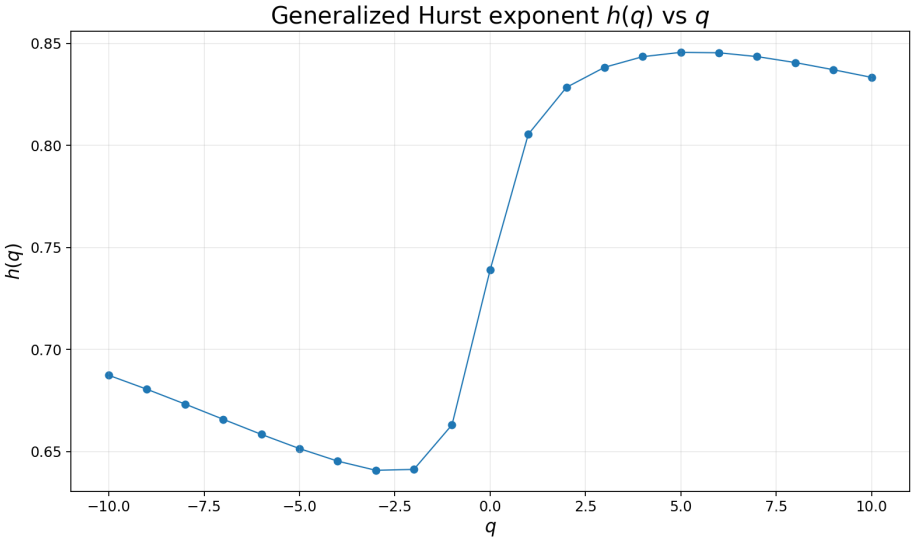
For each moving window:

$$h(q,t), \qquad \Delta h(t) = h(0,t) - h(5,t), \qquad \Delta\alpha(t) = \alpha_{\max}(t) - \alpha_{\min}(t).$$

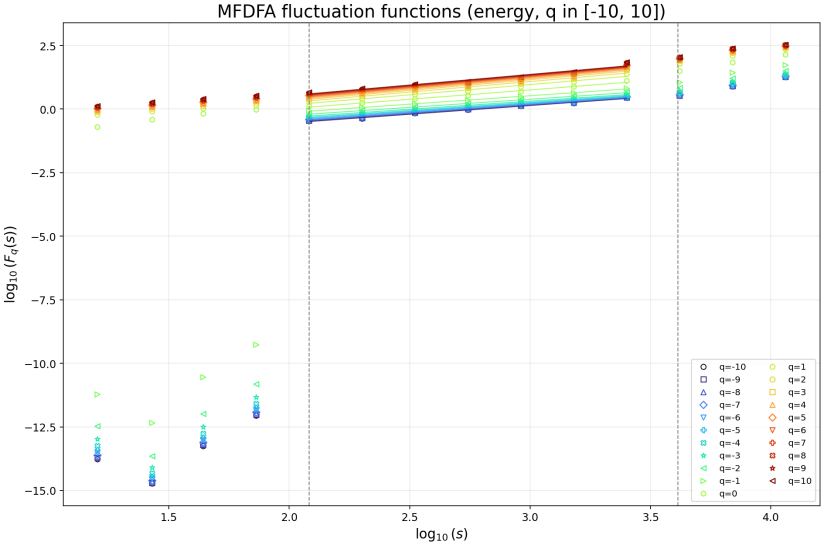
How to read the diagnostics

- $h(2)$: classical persistence indicator.
  - $h(2) \approx 0.5$ random,
  - $h(2) > 0.5$ persistent,
  - $h(2) < 0.5$ anti-persistent.
- $h(q)$ **vs** q: curvature indicates multifractality.
- $F_q(s)$: straight lines in log-log space indicate scaling; kinks indicate crossovers/regime changes.
- $f(\alpha)$: spectrum width $\Delta\alpha$ measures multifractality strength.
- $\Delta h(t)$ **and** $\Delta\alpha(t)$: time-resolved multifractality strength.

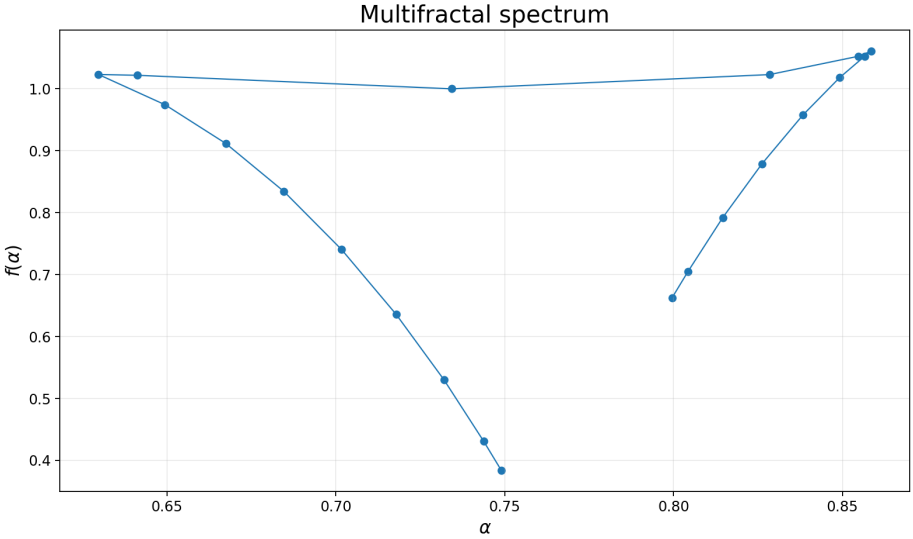
Generalized Hurst exponents $h(q)$



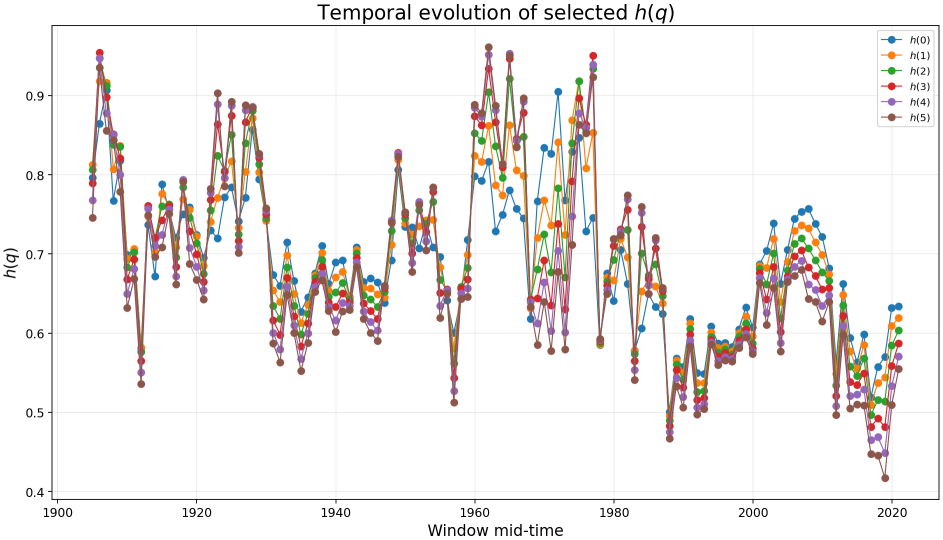
Fluctuation functions $F_q(s)$ and fitted scaling range



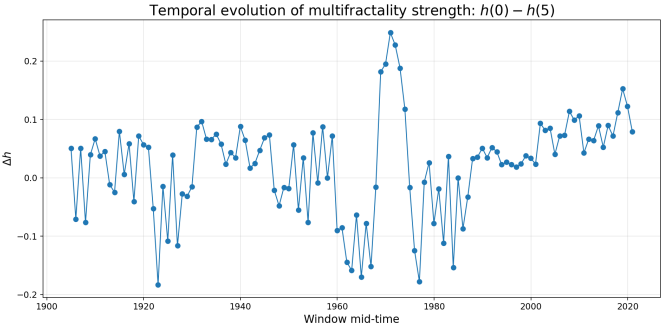
Multifractal spectrum $f(\alpha)$



Temporal evolution of selected $h(q)$



Multifractality strength over time: $\Delta h(t)$

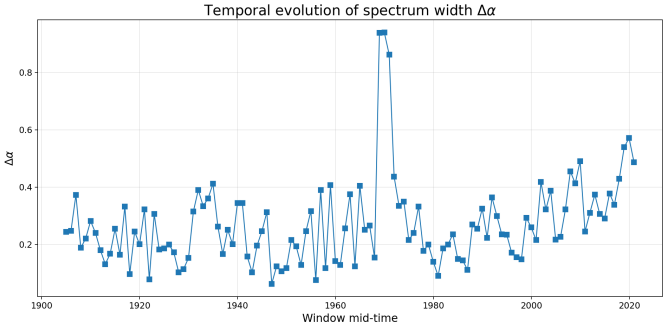


- Recommended robust definition for positive-q windows:

$$\Delta h(t) = h(0, t) - h(5, t).$$

- Larger values indicate stronger separation between typical fluctuations and bursts.
- Spikes may indicate enhanced clustering, intermittency, or regime changes.

Spectrum-width evolution over time: $\Delta\alpha(t)$

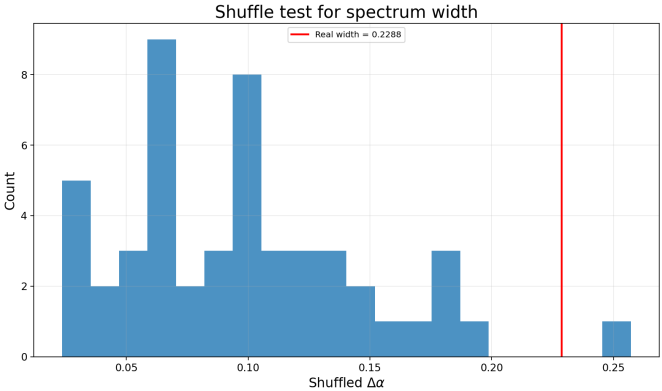


- Windowed spectrum width:

$$\Delta\alpha(t) = \alpha_{\max}(t) - \alpha_{\min}(t).$$

- Broad spectrum $\Rightarrow$ stronger multifractality.
- Compare directly with $\Delta h(t)$ to identify robust changes.

Shuffle test: real multifractality or ordering artifact?



- Shuffling preserves the value distribution but destroys temporal order.
- If the real spectrum width is much larger than shuffled widths, then temporal correlations matter.
- If real and shuffled widths are similar, multifractality is mostly distribution-driven.

Interpretation strategy for seismic catalogs

1. **Check $F_q(s)$ first**: only interpret scaling exponents in a linear range.
2. **Use moderate q-ranges**: $[-5, 5]$ or $[-10, 10]$ are usually more stable than $[-30, 30]$.
3. **Compare observables**: counts vs energy vs thresholded subcatalogs.
4. **Inspect time dependence**: use $h(q, t)$, $\Delta h(t)$, and $\Delta\alpha(t)$ to locate regime changes.
5. **Validate with surrogates**: shuffle test (already available), then optionally phase-randomized or IAAFT surrogates.

Main scientific takeaways

- Persistent scaling can be monitored through $h(2, t)$.
- Multifractality strength is captured compactly by:

$$\Delta h(t) = h(0, t) - h(5, t), \qquad \Delta\alpha(t) = \alpha_{\max}(t) - \alpha_{\min}(t).$$

- The combination of windowed MFDFA + surrogates provides a practical path toward:
  - regime classification,
  - comparison of full and thresholded catalogs,
  - testing whether apparent multifractality is correlation-driven.

Next steps

- Add **phase-randomized** and **IAAFT** surrogate tests.
- Compare raw vs declustered catalogs.
- Repeat for multiple bin sizes (1D, 3D, 7D).
- Compare counts and energy for the same windows.
- Examine whether spikes in $\Delta h(t)$ or $\Delta\alpha(t)$ align with major seismic episodes.

Thank you